How AYUSH/Ayurveda Is Handled — MediKiosk

Standalone reference covering the full Ayurvedic-specific architecture: history-taking, document handling, and terminology. This is the differentiation focus of the project — most competing teams will bolt on a shallow "Ayurveda mode"; this doc describes doing it structurally correctly instead.

---

1. Why Ayurveda needs different handling, not just extra questions

Ayurvedic history-taking is **person-centered, not just disease-centered** — two patients with the same complaint can need different treatment based on constitution. This is a structural difference from allopathic intake, not a vocabulary difference.

Dashavidha Pariksha (tenfold examination) — the ten parameters: Prakriti (constitution), Vikriti (current imbalance), Sara (tissue quality), Samhanana (structural integrity), Pramana (anthropometry), Satmya (adaptability), Satva (mental strength), Ahara

Shakti (digestive capacity), Vyayama Shakti (exercise capacity), Vaya (age).

2. Three distinct handling modes, not one uniform flow

Component	Nature	Handling
Prakriti	Fixed lifelong constitution, deterministic scoring	Frontend-only, multiple-choice, no LLM
Vikriti / Agni / Koshtha	Current-state, classified into known categories	LLM, but *constrained classification*, not open generation
Standard HPI / chief complaint	Same as allopathic	Existing SOCRATES-style dialogue manager, shared engine

2a. Prakriti — frontend-only, assessed once per patient (not per visit)

- Source: CCRAS Prakriti Assessment Scale (PAS) — Ministry of AYUSH (CCRAS)-validated instrument, 91 predictors (31 Vata / 29 Pitta / 32 Kapha), across four trait categories (physical, physiological, psychological, behavioral).

- **Scoring is rule-based counting, not LLM interpretation**: each predictor = 1 mark to whichever dosha it indicates (or no mark, if non-indicative). Dominant Prakriti = highest raw score(s).
- **Shortened for demo**: ~20 hand-picked predictors from the real 91 (spanning all four categories), sourced from the actual CCRAS manual/portal where possible. Fallback: Prakriti200 (24-item, CCRAS-guideline-aligned, open dataset) if real CCRAS access isn't obtained in time — disclose this substitution honestly if used.
- **Runs entirely client-side** — static question data + a scoring function in the frontend, no ASR/translation/LLM/backend round-trip per question. Only the final `{vata, pitta, kapha}` result is POSTed once.
- **Assessed once per patient, ever, not once per visit** — Prakriti is fixed at conception per Ayurvedic theory, so it doesn't need re-asking at every OPD visit. Attaches to the *patient* record, not the *visit* record. First-ever Ayurvedic visit runs the full flow; every subsequent visit just fetches the stored result. This is what keeps the "AYUSH

depth vs. speed" tension resolved — the slow part happens once per patient's lifetime, not per visit.

- Full detail: see `prakriti-questionnaire-implementation.md`.

2b. Vikriti / Agni / Koshtha — LLM-driven, constrained classification

- These reflect the patient's **current state** (unlike Prakriti), so must be assessed at every visit.
- Needs an LLM because patient answers are unstructured natural language (voice/touch, translated to English) — but the LLM's job is **classification into a small, fixed set of classical Ayurvedic categories**, not free-form follow-up generation like SOCRATES.
- Example categories: Agni → Sama (balanced) / Vishama (irregular, Vata-type) / Tikshna (sharp, Pitta-type) / Manda (slow, Kapha-type). Vikriti → which dosha's imbalance signs are present.
- Prompt pattern:

```
"Classify this patient's Agni into one of: Sama/Vishama/Tikshna/Manda,
 based on these classical criteria: [criteria list].
 Ask at most one clarifying question if ambiguous."
→ {"agni_type": "vishama",
```

```
"confidence": "medium", "reasoning":
"..."}
```

- **Requires domain research input**: someone needs to source the actual classical criteria (from Charaka Samhita descriptions or a modern Ayurvedic diagnostic reference) for what patient-reported signs indicate each category — the LLM should be grounded in real criteria, not left to freely guess at what "Vata-type digestive imbalance" looks like.
- More predictable than open SOCRATES branching (small fixed category set, at most 1-2 clarifying questions) but still needs LLM reasoning, unlike Prakriti.

2c. Dialogue manager mode split

```
IF history_type == "ayurvedic":
    1. Standard chief-complaint/HPI intake
(shared engine with allopathic)
    2. Prakriti — SKIP if already on file
for this patient; run once if not
(frontend-only)
    3. Vikriti/Agni/Koshtha — LLM,
constrained classification (every visit)
ELSE (allopathic):
    Standard SOCRATES-style adaptive
branching only
```

The critical distinction to get right: **Prakriti must NOT be complaint-reactive.** A patient's constitution doesn't change because they came in with a headache instead of a stomach ache. Treating Prakriti with the same adaptive-follow-up logic as SOCRATES/Vikriti is a subtle but real design error — most competing teams that bolt on "Ayurveda mode" get exactly this wrong.

3. Mixed-system visits (integrative care)

Real hospitals (including AIIA) sometimes deliver integrative care in one visit. Handling:

```
history_type: "allopathic" | "ayurvedic" |
"mixed"

IF "mixed":
    - Patient/staff selects PRIMARY
framework at check-in
    - Run the full PRIMARY framework's
interview
    - Run a SHORT supplementary pass for
the secondary framework —
      only the non-overlapping piece (e.g.,
if primary=allopathic,
      secondary=ayurvedic: just the
Prakriti/Vikriti/Agni block,
      since HPI is already covered)
```

Never run two full parallel interviews — the overlap (chief complaint/HPI) is asked once regardless of framework; only framework-specific, non-overlapping content gets added.

Scoping note: build pure allopathic and pure ayurvedic flows first (core demo story); add "mixed" as a thin extension once those are stable, since the underlying components already exist — mixed mode mostly means "run both blocks in one visit."

4. Cross-system context — past records inform, don't duplicate, today's questions

If a patient has records from the *other* system (e.g., an old allopathic lab report, seen today for an Ayurvedic visit), it should sharpen today's single interview, not trigger a second one:

```
Past allopathic lab: HbA1c 8.2% (diabetes-range)
  → fed into today's Ayurvedic dialogue manager as context:
    "Patient has a prior biomedical diagnosis suggestive of impaired
     glucose metabolism. When assessing Agni and Ahara-Vihara, probe
     specifically around this."
  → Same Ayurvedic-framework questions
```

```
asked, just better-informed —
    NOT a second parallel allopathic
interview
```

Both systems' diagnoses can coexist in the final output JSON (current visit's Ayurvedic diagnosis + referenced prior diagnoses from other systems) — that's where "both frameworks" genuinely belongs, not in the live conversation.

5. Document schema — shared, with Ayurveda-aware extension (not a separate schema)

Prescriptions/lab reports/discharge summaries share the same schema shape regardless of medicine system — only vocabulary differs:

```
{
  "medicine_system": "allopathic |
ayurvedic | unani | other",
  "medications": [
    {
      "name": "string",
      "dosage": "string",
      "frequency": "string",
      "form": "string | null"   // churna,
kwath, vati, taila, arishta, asava, ghrita
    }
  ],
```

```
  "diagnosis": [
    {
      "condition": "string",
      "system_terminology": "ayurvedic | biomedical",
      "coding_system": "NAMASTE | ICD | null",
      "code": "string | null"
    }
  ]
}
```

OCR/extraction challenges specific to Ayurvedic documents: mixed English/Devanagari/Sanskrit script in one document; non-Western formulation names (Triphala, Ashwagandha, Chyawanprash) less represented in OCR training data → expect more low-confidence flags on these terms specifically. Extraction prompt (ML4) needs explicit awareness it may be reading an Ayurvedic document, with example formulation names/forms provided.

6. NAMASTE — terminology coding for Ayurvedic diagnoses

- **NAMASTE** (National AYUSH Morbidity & Standardized Terminologies Electronic Portal) — Ministry of AYUSH's standardized coding system for Ayurveda/Siddha/Unani diagnoses, analogous to ICD for biomedicine.

- Mapped to WHO ICD-11 Chapter 26, Module 2 (TM2) — officially released Feb 2025, so this is live infrastructure, not a proposal.
- This is the intended real path for FHIR/ABDM semantic interoperability on the Ayurvedic side — using it (rather than leaving Ayurvedic diagnoses as unstructured text) is the officially-correct approach, not a workaround.
- Open-source bridge available: **AyushSyncAPI** — FHIR R4-compliant terminology microservice with bidirectional NAMASTE ↔ ICD-11 TM2 mapping, auto-complete search endpoints, ABHA auth built in. Worth evaluating for reuse instead of building terminology lookup from scratch.
- **Caveat**: public access to the full official NAMASTE dataset has been inconsistent in past implementation attempts — verify direct access before committing; have a small representative code sample ready as fallback for the demo if full access isn't available.

7. Judges-facing framing

- "We didn't bolt Ayurveda questions onto our allopathic engine — Prakriti and Vikriti/Agni are structurally different from SOCRATES branching, and from each other, because they *are* structurally different things in Ayurvedic diagnostics."

- "Prakriti uses the actual CCRAS-validated instrument's predictors and scoring logic, assessed once per patient, not once per visit — which is both clinically correct and what makes it fast."
- "Ayurvedic diagnoses are NAMASTE-coded for real FHIR/ABDM interoperability, not left as unstructured text."
- Honest caveats to state plainly, not hide: CCRAS-PAS is CCRAS-validated but not free of scientific-validity debate in the wider literature (per the critical review — only 2 of 64 known Prakriti tools show methodological rigor); demo uses a subset of the full 91-item scale for time; NAMASTE dataset access may be partial/sample-based for the demo.